

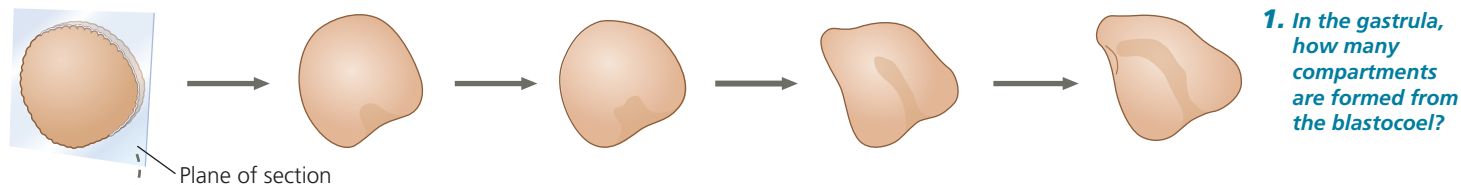
The embryos of all animals, and only animals, gastrulate. Cells change position: Some are brought inside the embryo, and others spread over the surface. The net result is an embryo with two, or more commonly three, cell layers. This figure will help you visualize the basic choreography of gastrulation before you explore the specific steps in different types of animals.

➔ **Instructors:** Additional questions related to this Visualizing Figure can be assigned in **Mastering Biology**.

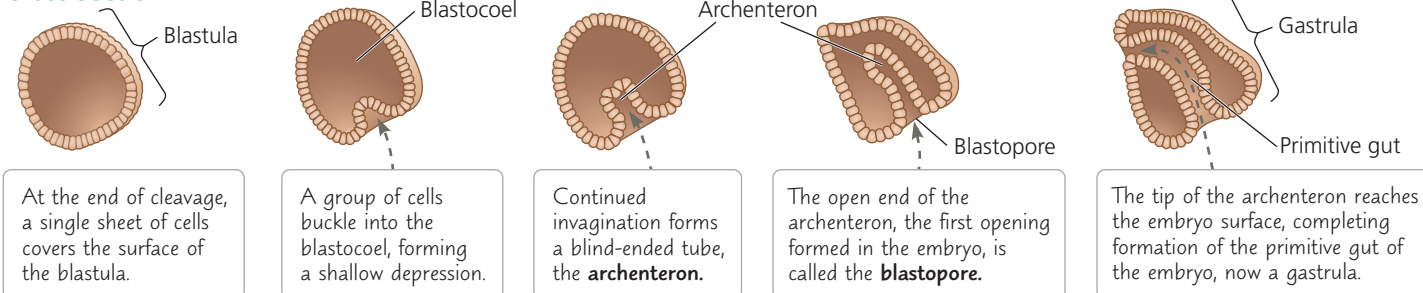
Reorganizing the animal embryo in three dimensions

Gastrulation typically begins with invagination, the infolding of a sheet of cells, shown here in both surface view and cross section. The resulting changes to the epithelium covering the embryo resemble what happens if you push a finger into one end of a lightly inflated balloon.

Surface view:



Cross section:



Forming the primary cell layers of the animal body

In diploblasts, gastrulation forms two germ layers: ectoderm and endoderm. Triploblasts—both protostomes and deuterostomes—gain a third layer: mesoderm. At the end of embryogenesis, each germ layer gives rise to specific tissues and organs. To visualize how this occurs, trace the fate of each germ layer in each row of drawings (all views are cross sections except the sea urchin larva, which is a transparent surface view).

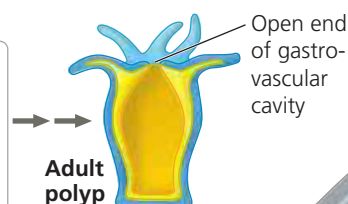
Key

- Ectoderm
- Mesoderm
- Endoderm

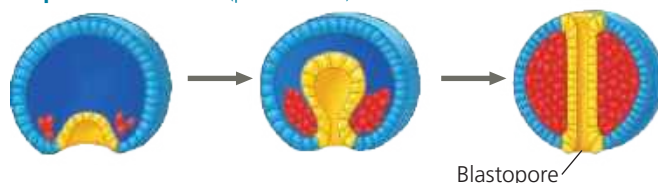
Diploblast: *Hydra*



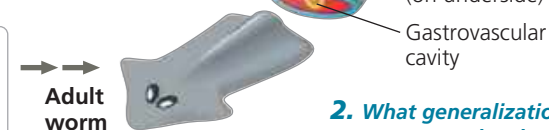
In diploblasts, the blastopore becomes the open end of the gastrovascular cavity.



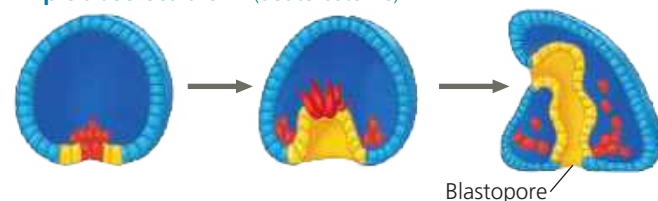
Triploblast: *Planaria* (protostome)



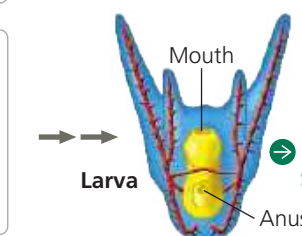
In protostomes, the mouth forms from the blastopore.



Triploblast: Sea urchin (deuterostome)



In deuterostomes, including all vertebrates as well as some invertebrates, the mouth forms *opposite* the blastopore.



2. What generalizations can you make about the location of mesoderm, when it is present, during and after development?

➔ **Mastering Biology Video:** Sea Urchin Gastrulation